STAT 20000 Chapter 10
The Regression Method

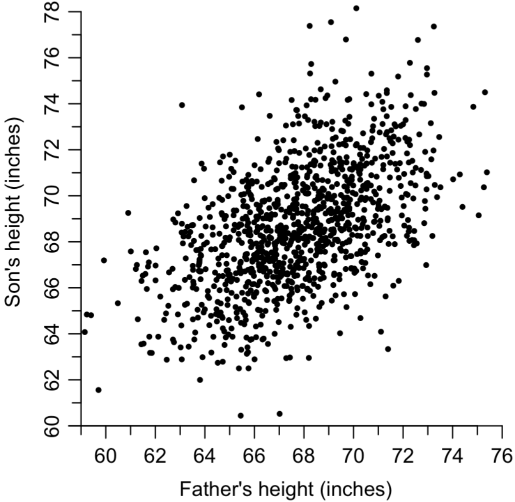
Outline

- Regression analysis is used to describe the distribution of values of one variable, the **response**, as a function of other – **explanatory** variables.
  - STAT20000 deals with *simple* linear regression only, where there is only one explanatory variable.
- Regression line *vs* SD line.
- The regression fallacy.
- Given the scatter diagram of x and y values, there are **two** regression lines
  - Regression of y on x;
  - Regression of x on y.

Pearson's Father-Son Data

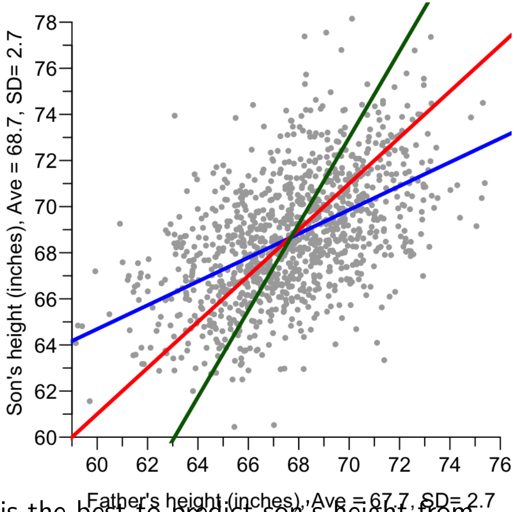
The scatter plot below shows the heights of 1078 fathers and their full-grown sons, in England, circa 1900. There is one dot for each father-son pair.

- The plot shows a moderate linear relationship between fathers' heights and the sons' heights
- For a father of a given height, what height would you predict for his son?



The SD Line & The Regression Line

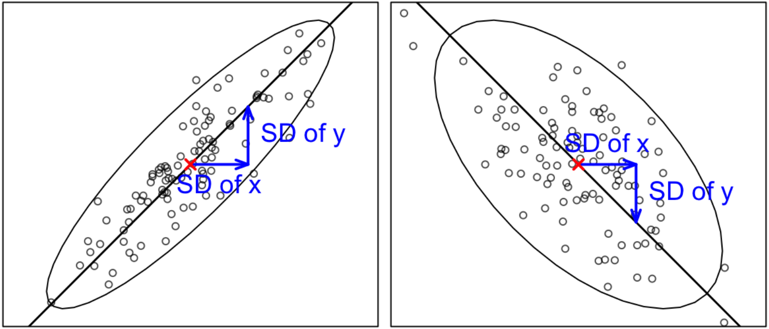
- Red: the *SD line*.
- Blue: the *regression line* to predict the son's height from from his father's height
- We will introduce the Green line shortly.



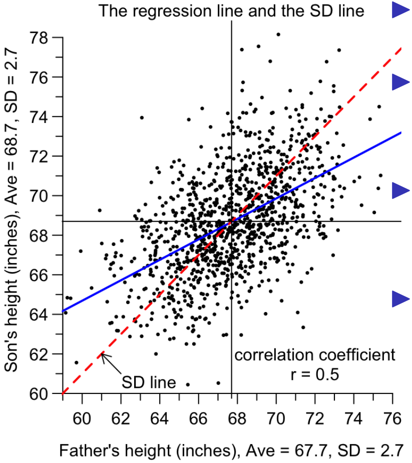
Which line do you think is the best to predict son's height from father's height?

The SD Line

- The SD line goes through the point of averages $(\bar{x}, \bar{y})$.
- When $\begin{matrix} r > 0 \\ r < 0 \end{matrix}$, the line $\begin{matrix} \text{rises} \\ \text{falls} \end{matrix}$ one vertical SD for each run of one horizontal SD.
- The SD line is about the same as the long axis of the oval which encircle the main portion of the cloud of points.

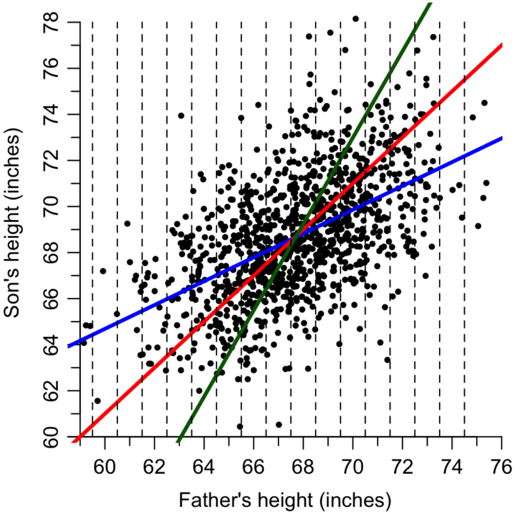


Regression Line v.s. SD Line



- Both lines go through the point of averages $(\bar{x}, \bar{y})$.
- The regression line is *less steeply sloped* than the SD line, by a factor of r.
- If $r > 0$, whenever X increases by 1 (SD of X), Y only increase of only r (SD of Y)
- If $r < 0$, whenever X increases by 1 (SD of X), Y only decrease of only r (SD of Y)

The points in the vertical chimney are the father-son pairs where the father is 72 inches tall, to the nearest inch.



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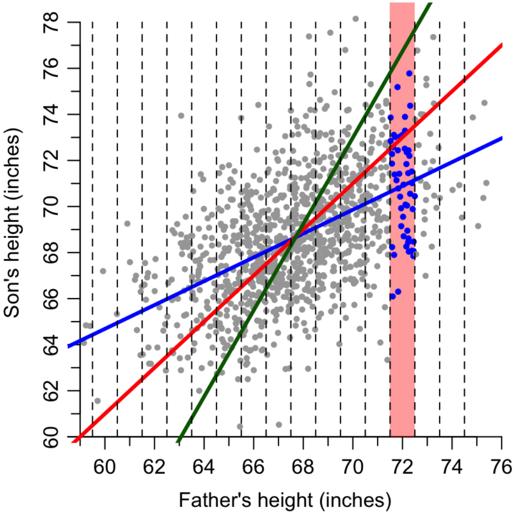
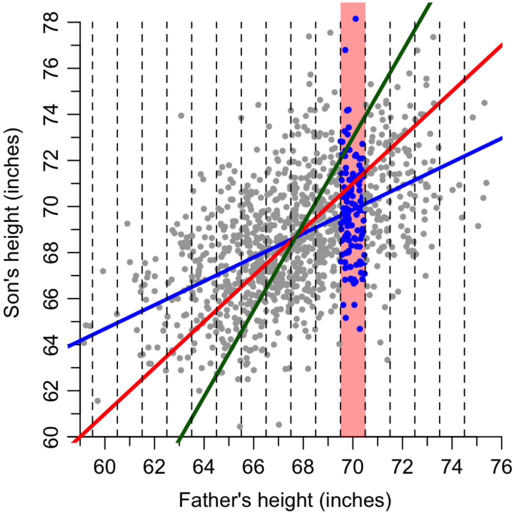
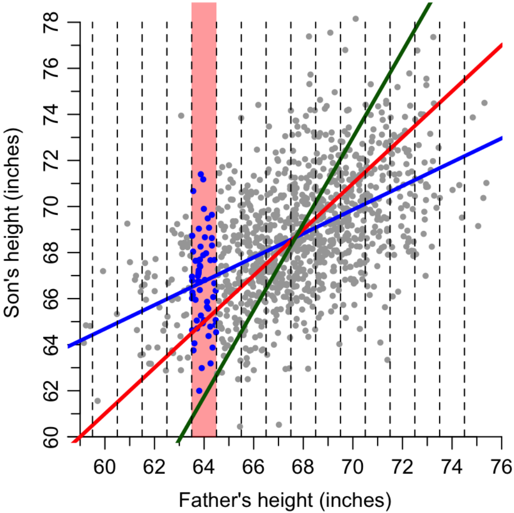


The points in the vertical chimney are the father-son pairs where the father is 70 inches tall, to the nearest inch.



The points in the vertical chimney are the father-son pairs where the father is 64 inches tall, to the nearest inch.



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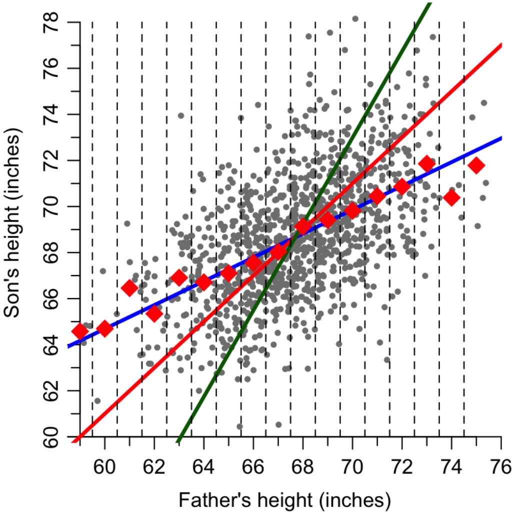
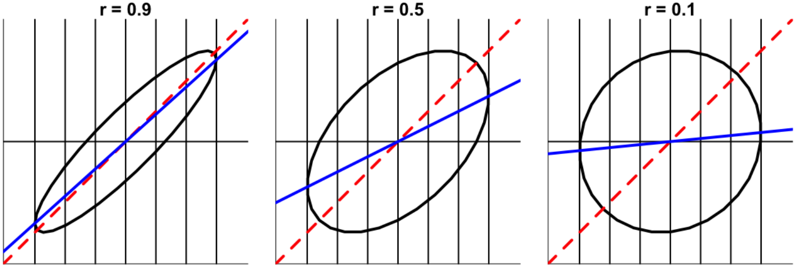


Why the Regression Line $\neq$ the SD line?

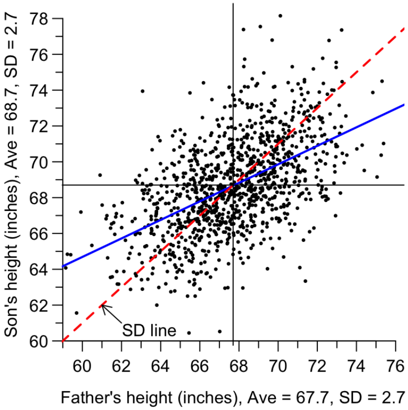
For a football-shaped scatter diagram, as the football is cut into vertical strips, the midpoints of the vertical strips falls on the regression line, not the SD line



- SD line is used to describe the shape of the scatter diagram, not for prediction.
- When $r = 1$ or $r = -1$, the two lines coincide.

Where Does the Term “Regression” Come From?

Sons of fathers who are taller / shorter than average are themselves taller / shorter than average, but by not so much as the fathers.



- There is a "falling back"towards the mean. In Galton’s terms: a regression towards mediocrity."
- The term "regression" is widely used in statistics in connection with how one variable varies with respect to another.

Q: Why do sons of fathers who are taller than average tend themselves to be taller than average, but by not so much as their fathers?

Q: Why do sons of fathers who are taller than average tend themselves to be taller than average, but by not so much as their fathers?

- Their mothers might not be as exceptionally tall as their fathers.

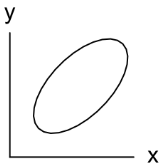
Q: Why do sons of fathers who are taller than average tend themselves to be taller than average, but by not so much as their fathers?

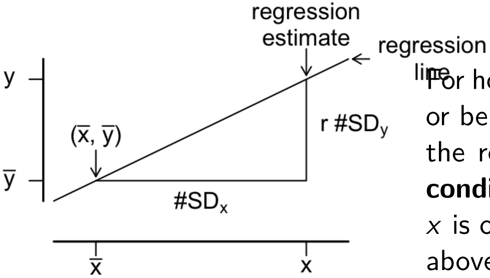
- Their mothers might not be as exceptionally tall as their fathers.

- Diet, exercise, and other environmental factors also influence height. It is more common to get an average level of diet, exercise, and other factors than to get exceptional level of these factors.

The Regression Method

For any football-shaped scatterplot (i.e., the majority of the points in the plot are distributed in a sloped oval), the average value of y corresponding to a given value of x may be estimated by the regression line (for y on x).





For however many SDs x is above or below $\bar{x}$,
the regression estimate for **the conditional average of y given x** is only r times that many SDs above or below $\bar{y}$.

Whenever x goes up by 1 SD_x, y goes up by r SD_y, on average.

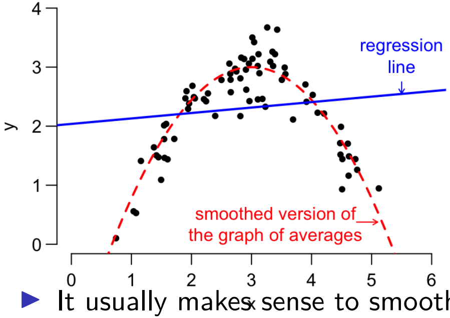
Why is r the Right Factor?

Short answer: it just turns out that way.
It happens that way for all football-shaped scatted diagrams, e.g. Pearson's father-son data.

Long answer:
Three special cases are easy to understand: $r = 0$, 1, and -1.

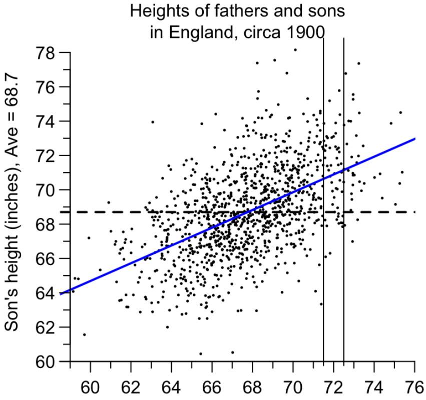
- If $r = 0$, there is no association between x and y. For a one SD increase in x, there is no increase or decrease in y, on average.
- If $r = 1$, there is a perfect positive association between x and y. A one SD increase in x corresponds exactly to a one SD increase in y.
- If $r = -1$, there is a perfect negative association. A one SD increase in x corresponds exactly to a one SD decrease in y.

Nonlinear Association



- **Beware**: linear regression doesn't make sense when the data structure isn't linear.
- You can always use the graph of averages to describe how y changes with respect to x, on average.
- It usually makes sense to smooth the graph of averages to get a simpler description of the variation.
- But when the pattern in the data is nonlinear, it doesn't make sense to smooth the graph of averages to a straight line.
  - How do you tell if pattern in the data is nonlinear?

Predictions for Individuals

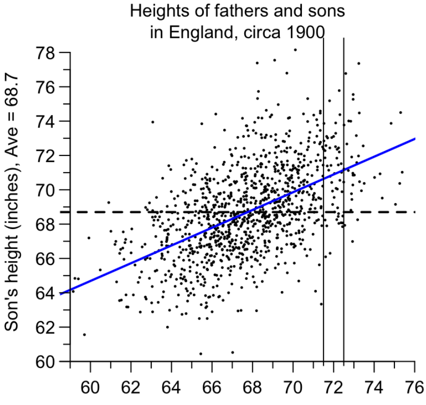


Consider again the 1,078 pairs of fathers and sons.

I'm going to pick one of these pairs at random, and ask you to predict the son's height.

- If I tell you nothing about the father, your prediction would be ______________. In the diagram, that corresponds to using the ________________ line.
- If I tell you the father's height, your prediction would be ______________________________. In the diagram, that corresponds to using the ________________.

Predictions for Individuals



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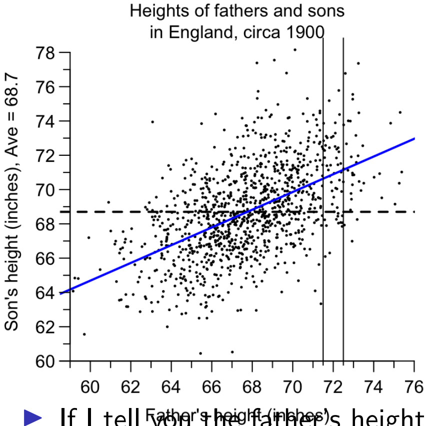
I'm going to pick one of these pairs at random, and ask you to predict the son's height.

- If I tell you nothing about the father, your prediction would be <u>68.7 inches</u>. In the diagram, that corresponds to using the <u>horizontal dash</u> line.

- If I tell you the father's height, your prediction would be ________________________. In the diagram, that corresponds to using the ________________.

Predictions for Individuals



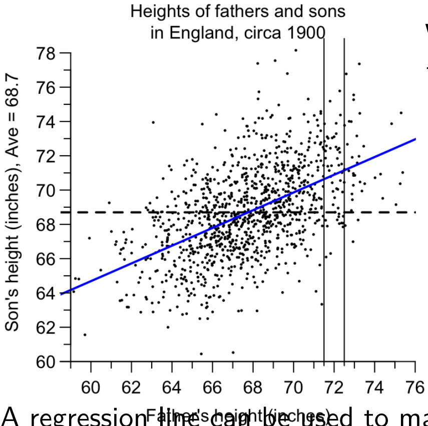
Consider again the 1,078 pairs of fathers and sons.

I'm going to pick one of these pairs at random, and ask you to predict the son's height.

- If I tell you nothing about the father, your prediction would be 68.7 inches. In the diagram, that corresponds to using the horizontal dash line.

- If I tell you the father's height, your prediction would be ave ht of sons w/ fathers of that ht. In the diagram, that corresponds to using the regression line.

Be Cautious for Extrapolation



Would you use the regression line to predict the height of a son:

- For fathers in England circa 1900?
- For fathers in England circa 1900 who were extremely short/tall?
- For fathers in the U.S. in 2011?

A regression line can be used to make predictions for individuals. But if you have to extrapolate far from the data, or to a different group of subjects, watch out!

Using the Regression Line

Consider the father-son data. The scatter plot is football-shaped.

$$\text{average height of fathers} \approx 68\text{''},\ \text{SD} \approx 3\text{''}$$
$$\text{average height of sons} \approx 69\text{''},\ \text{SD} \approx 3\text{''},\ \ r \approx 0.5$$

On average, how tall are sons of 6 foot fathers?

- A 6 foot father is _____ inches above average in height.
- That's _______ of an SD.
- The average height of sons of 6 foot fathers is _________ the overall average by __________________ of an SD.
- That's _____ inches.
- So sons of 6 foot fathers are on average ______________ inches tall.

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On average, how tall are sons of 6 foot fathers?

- A 6 foot father is __4__ inches above average in height.
- That's _______ of an SD.
- The average height of sons of 6 foot fathers is _________ the overall average by __________________ of an SD.
- That's _____ inches.
- So sons of 6 foot fathers are on average ______________ inches tall.

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On average, how tall are sons of 6 foot fathers?

- A 6 foot father is __4__ inches above average in height.
- That's $\underline{\ \ 4/3\ \ }$ of an SD.
- The average height of sons of 6 foot fathers is __above__ the overall average by ________________ of an SD.
- That's _____ inches.
- So sons of 6 foot fathers are on average ________________ inches tall.

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On average, how tall are sons of 6 foot fathers?

- A 6 foot father is __4__ inches above average in height.
- That's __$4/3$__ of an SD.
- The average height of sons of 6 foot fathers is __above__ the overall average by __$0.5 \times 4/3 = 2/3$__ of an SD.
- That's _____ inches.
- So sons of 6 foot fathers are on average _______________ inches tall.

Using the Regression Line

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- That's __$4/3$__ of an SD.
- The average height of sons of 6 foot fathers is __above__ the overall average by __$0.5 \times 4/3 = 2/3$__ of an SD.
- That's __2__ inches.
- So sons of 6 foot fathers are on average __________ inches tall.

Using the Regression Line

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$$\text{average height of fathers} \approx 68\text{''},\ \text{SD} \approx 3\text{''}$$

$$\text{average height of sons} \approx 69\text{''},\ \text{SD} \approx 3\text{''},\ \ r \approx 0.5$$

On average, how tall are sons of 6 foot fathers?

- A 6 foot father is __4__ inches above average in height.
- That's __$4/3$__ of an SD.
- The average height of sons of 6 foot fathers is __above__ the overall average by __$0.5 \times 4/3 = 2/3$__ of an SD.
- That's __2__ inches.
- So sons of 6 foot fathers are on average __$69 + 2 = 71$__ inches tall.

Using the Regression Line (Cont'd)

Below is a summary for the height and systolic blood pressure for the men ages 18–24 in the HANES sample.

$$\begin{aligned} \text{average height} &\approx 68\text{"}, & \text{SD} &\approx 3\text{"} \\ \text{average blood pressure} &\approx 124\text{mm}, & \text{SD} &\approx 15\text{mm}, \quad r \approx -0.2 \end{aligned}$$

The scatter diagram is football-shaped. Estimate the average blood pressure of 64-inch tall men.

Step 1. Convert $x = 64$ into standard units:

Step 2. Multiply the SU in Step 1 by $r = -0.2$. The product is the predicted y-value in standard units.

Step 3. Convert the standard unit of the predicted y SU_y back to the original scale.

So the average blood pressure of these men is estimated as _______ mm.

Using the Regression Line (Cont'd)

Below is a summary for the height and systolic blood pressure for the men ages 18–24 in the HANES sample.

$$\begin{array}{rll} \text{average height} \approx 68", & \text{SD} \approx 3" & \\ \text{average blood pressure} \approx 124\text{mm}, & \text{SD} \approx 15\text{mm}, & r \approx -0.2 \end{array}$$

The scatter diagram is football-shaped. Estimate the average blood pressure of 64-inch tall men.

Step 1. Convert $x = 64$ into standard units:
$(\text{SU of } x) = (64 - 68)/3 = -4/3$
Step 2. Multiply the SU in Step 1 by $r = -0.2$. The product is the predicted y-value in standard units.

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So the average blood pressure of these men is estimated as ______ mm.

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$$r \times (\text{SU of } x) = (-0.2) \times (\frac{-4}{3}) = \frac{4}{15} = (\text{SU of } y)$$

Step 3. Convert the standard unit of the predicted y SU_y back to the original scale.

So the average blood pressure of these men is estimated as _______ mm.

Using the Regression Line (Cont'd)

Below is a summary for the height and systolic blood pressure for the men ages 18–24 in the HANES sample.

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$$r \times (\text{SU of } x) = (-0.2) \times (\frac{-4}{3}) = \frac{4}{15} = (\text{SU of } y)$$

Step 3. Convert the standard unit of the predicted y SU_y back to the original scale.

predicted $y = (\text{Ave of } y) + (\text{SU of } y)(\text{SD of } y) = 124 + (4/15) \times 15 \approx 128.$

So the average blood pressure of these men is estimated as _______ mm.

Using the Regression Line (Cont'd)

Below is a summary for the height and systolic blood pressure for the men ages 18–24 in the HANES sample.

$$\begin{aligned} \text{average height} &\approx 68", & \text{SD} &\approx 3" \\ \text{average blood pressure} &\approx 124\text{mm}, & \text{SD} &\approx 15\text{mm}, \quad r \approx -0.2 \end{aligned}$$

The scatter diagram is football-shaped. Estimate the average blood pressure of 64-inch tall men.

Step 1. Convert $x = 64$ into standard units:
$(\text{SU of } x) = (64 - 68)/3 = -4/3$
Step 2. Multiply the SU in Step 1 by $r = -0.2$. The product is the predicted y-value in standard units.

$$r \times (\text{SU of } x) = (-0.2) \times (\frac{-4}{3}) = \frac{4}{15} = (\text{SU of } y)$$

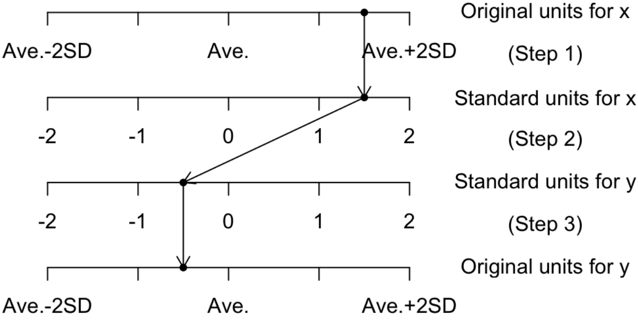
Step 3. Convert the standard unit of the predicted y SU_y back to the original scale.

predicted $y = (\text{Ave of } y) + (\text{SU of } y)(\text{SD of } y) = 124 + (4/15) \times 15 \approx 128.$

So the average blood pressure of these men is estimated as __128__ mm.

In general, to predict the value of of y for a given value of x, you:

(1) convert the given value of x to **standard units**,
(2) multiply by the **correlation (r)**, which gives the corresponding average value of y in standard units.
(3) convert back from standard units.



In Step 1, you use the Average and SD of x .
In Step 3, you use the Average and SD of y .

In-Class Exercise

For the men age 18 and over in HANES5,

$$\begin{array}{lll} \text{average height} \approx 69 \text{ inches}, & \text{SD} \approx 3 \text{ inches} & \\ \text{average weight} \approx 190 \text{ lbs}, & \text{SD} \approx 42 \text{ lbs}, & r = 0.41 \end{array}$$

a. Predict the weight of a 70.8-inch tall man.
b. Predict the weight of a 64.5-inch tall man.

Answer to Exercise

average height $\approx$ 69 inches, SD $\approx$ 3 inches

average weight $\approx$ 190 lbs, SD $\approx$ 42 lbs, $r = 0.41$

a. Predict the weight of a 70.8-inch tall man.

- Step 1. Convert $x = 70.8$ into standard units:

$$(\text{SU of } x) = \frac{70.8 - 69}{3} = 0.6$$

- Step 2. Multiply the SU in Step 1 by $r = 0.41$. The product is the predicted y-value in standard units.

$$r \times (\text{SU of } x) = 0.41 \times 0.6 = 0.246 = (\text{SU of } y)$$

- Step 3. Convert the standard unit of the predicted y back to the original scale.

predicted $y = (\text{Ave of } y) + (\text{SU of } y)(\text{SD of } y) = 190 + 0.246 \times 42 \approx 200.33.$

So 70.8-inch men on average weigh 200.33 lbs.

Answer to Exercise

average height ≈ 69 inches, SD ≈ 3 inches

average weight ≈ 190 lbs, SD ≈ 42 lbs, $r = 0.41$

b. Predict the weight of a 64.5-inch tall man.

- Step 1. Convert $x = 64.5$ into standard units:

$$(\text{SU of } x) = \frac{64.5 - 69}{3} = -1.5$$

- Step 2. Multiply the SU in Step 1 by $r = 0.41$. The product is the predicted y-value in standard units.

$$r \times (\text{SU of } x) = 0.41 \times (-1.5) = -0.615 = (\text{SU of } y).$$

- Step 3. Convert the standard unit of the predicted y back to the original scale.

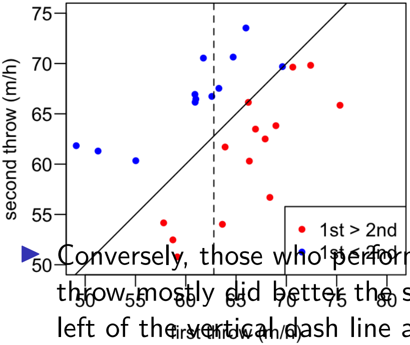
predicted $y = (\text{Ave of } y) + (\text{SU of } y)(\text{SD of } y) = 190 + (-0.615) \times 42 = 164$

So 64.5-inch men on average weigh 164.17 lbs.

Test-Retest Scenario, Regression Effect, and Regression Fallacy

The Test-Retest Scenario

Each pitcher in a baseball team made one throw, the coach commented on his performance before he made another throw.

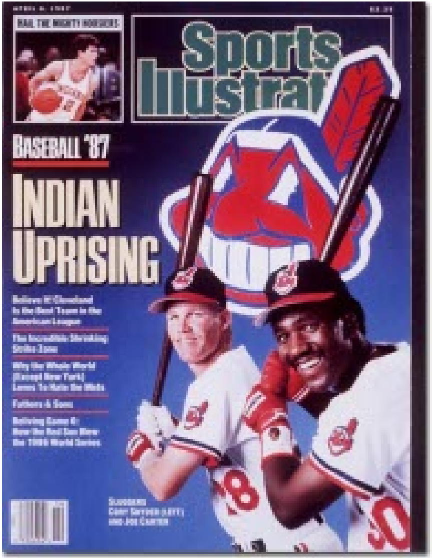


- vertical dash line marks the average of the 1st throw
- Those who did above average in the first throw tended to do worse the second time (most points to the right of the vertical dash line are red)
- Conversely, those who performed below the average in the first throw mostly did better the second time (most points to the left of the vertical dash line are blue)
- Does criticism help the pitchers while praise spoils them?

The Regression Effect and the Regression Fallacy

- In a typical test-retest situation, the subjects get different scores on the two tests. Take the bottom group on the first test. Some improve on the second test, others do worse; but on the average the bottom group shows an improvement. Now the top group. Some do better the second time, others fall back; but on average the top group does worse the second time. This is the **regression effect**, and it happens whenever the scatter diagram spreads out around the SD line into a football-shaped cloud of points.
- The **regression fallacy** consists in thinking that the regression effect is due to something other than spread around the SD line.

Example: Sports Illustrated Cover Jinx



The April 6, 1987 *Sports Illustrated* "Indian Uprising" cover.

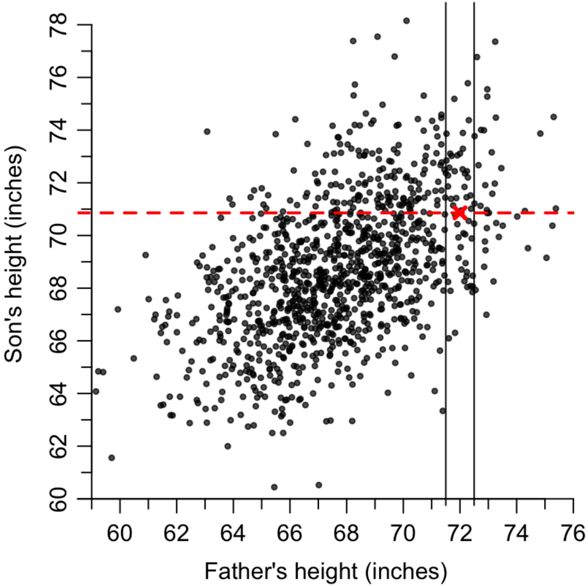
Individuals or teams who appear on the cover of the Sports Illustrated magazine will subsequently be jinxed.

In 1987, the Indians won 61 and lose 101 games, the worst of any team that season.

Not due to the jinx, but the regression effect.

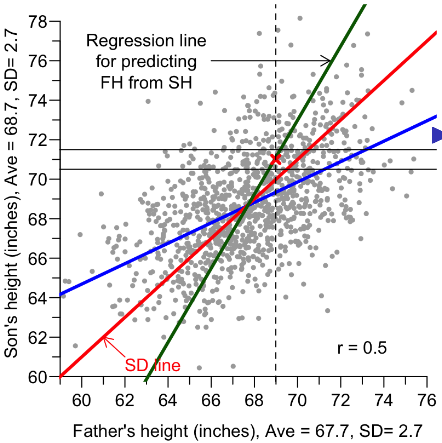
There Are Two Regression Lines

On p.16, we predicted that sons of 72-inch tall fathers are on average 71-inch tall. **True or False**: fathers of 71-inch tall sons are on average 72-inch tall.



There Are Two Regression Lines

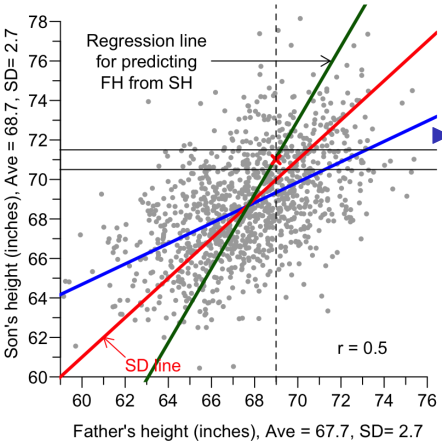
As well as regressing SH on FH, we can also regress FH on SH:



- Points in the horizontal strip are the father-son pairs where the son is 71" tall, to the nearest inch.
- The cross at the point $(68.99, 71)$ marks the average height of the fathers of these sons. The cross lies very close to the regression line for FH on SH, which predicts these fathers to be only __________, not 72".

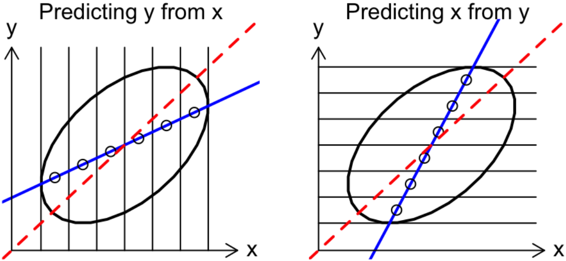
There Are Two Regression Lines

As well as regressing SH on FH, we can also regress FH on SH:



- Points in the horizontal strip are the father-son pairs where the son is 71" tall, to the nearest inch.
- The cross at the point $(68.99, 71)$ marks the average height of the fathers of these sons. The cross lies very close to the regression line for FH on SH, which predicts these fathers to be only <u>68.99"</u>, not 72".

Comparison of the Two Regression Lines



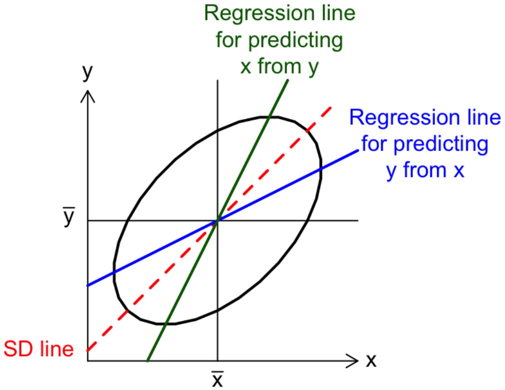
The regression line for predicting $\begin{array}{c} y \text{ from } x \\ x \text{ from } y \end{array}$ involves:

- dividing the scatter diagram into $\begin{array}{c}\text{vertical} \\ \text{horizontal}\end{array}$ strips;
- finding the average value of $\begin{array}{c} y \\ x \end{array}$ in each strip; and
- smoothing the resulting graph of averages to a straight line.

The line goes through the point of averages and is $\begin{array}{c}\text{less} \\ \text{more}\end{array}$ steeply sloped than the SD line by a factor of $\begin{array}{c} |r| \\ 1/|r| \end{array}$

There Are Two Regression Lines (Cont'd)

Two regression lines can be drawn across a scatter diagram:

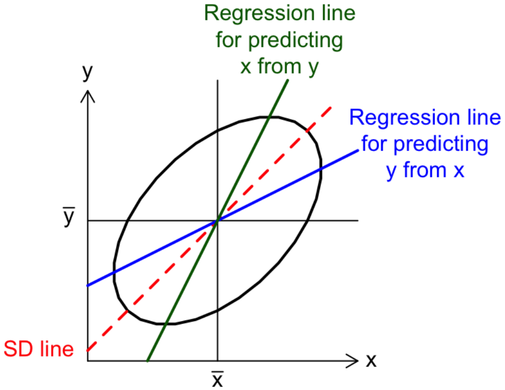


In general the two regression lines are different. You can screw up a prediction *badly* by using the wrong line!

- When will the two lines be the same?

There Are Two Regression Lines (Cont'd)

Two regression lines can be drawn across a scatter diagram:



In general the two regression lines are different. You can screw up a prediction *badly* by using the wrong line!

- When will the two lines be the same? When $r = 1$ or -1.

Summary

- Associated with each increase of one SD in x there is an increase of only r SDs in y, on the average. Plotting these *regression estimates* gives the *regression line* of y on x.
- The *graph of averages* is often close to a straight line, but may be a little bumpy. The regression line smooths out the bumps. If the graph of averages is a straight line, then it coincides with the regression line.
- The regression line can be used to make predictions of one variable from another. But if you have to extrapolate far from the data, or to a different group of subjects, be careful.
- Beware of nonlinear structure and outliers. They can fool the regression method.
- Remember the regression effect and the regression fallacy.
- You can draw two different regression lines on a scatter diagram: one predicts y from x; the other predicts x from y.